No.

Date

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B.

1). Pers : $f(x) = x^2 - x - 2 = 0$

$g(x) = x^2 - 4 = 0$

$$xr = b - \frac{f(b)(b-a)}{f(b) - f(a)}$$

hitung pendekatan untuk nilai x sampai iterasi ketiga metode regulafalsi

(1) $f(x) = x^2 - x - 2$

$a = 1$

$b = 3$

$f(1) = 1^2 - 1 - 2 = -2 \rightarrow f(a)$

$f(3) = 3^2 - 3 - 2 = 4 \rightarrow f(b)$

• iterasi 1 :

$$xr = 3 - \frac{(4(3-1))}{(4-(-2))}$$

$$xr = 3 - \frac{(4 \cdot 2)}{6}$$

$$xr = 3 - \frac{8}{6} = 1.667$$

• iterasi 2 :

$a = 1.667$

$b = 3$

$f(1.667) = 1.667^2 - 1.667 - 2 = -0.889$

$$xr = 3 - \frac{(4(3-1.667))}{(4-(-0.889))}$$

$$xr = 3 - \frac{(4 \cdot 1.333)}{4.889} = 1.909$$

• iterasi 3 :

$a = 1.909$

$b = 3$

$f(1.909) = 1.909^2 - 1.909 - 2 = -0.264$

$$xr = 3 - \frac{(4(3-1.909))}{(4-(-0.264))}$$

$$xr = 3 - \frac{(4 \cdot 1.091)}{4.264} = 1.977$$

Dipindai dengan CamScanner

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(2) $g(x) = x^2 - 4$

$a = 0$

$b = 3$

$g(0) = 0^2 - 4 = -4$

$g(3) = 3^2 - 4 = 5$

• Iterasi 1 :

$$xr = 3 - \frac{(5(3-0))}{(5-(-4))}$$

$$xr = 3 - \frac{(5 \cdot 3)}{9}$$

$$xr = 3 - \frac{15}{9} = 1{,}333$$

• Iterasi 2 :

$a = 1.333$

$b = 3$

$g(1{,}333) = 1{,}333^2 - 4 = -2{,}222$

$$xr = 3 - \frac{(5(3-1{,}333)}{(5-(-2{,}222))}$$

$$xr = 3 - \frac{(5 \cdot 1{,}667)}{7{,}222}$$

$$xr = 3 - \frac{8{,}335}{7{,}222} = 1{,}846$$

• Iterasi 3 :

$a = 1{,}846$

$b = 3$

$g(1{,}846) = 1{,}846^2 - 4 = -0{,}593$

$$xr = 3 - \frac{(5(3-1{,}846))}{(5-(-0{,}593))}$$

$$xr = 3 - \frac{(5 \cdot 1{,}154)}{5{,}593}$$

$$xr = 3 - \frac{5{,}770}{5{,}593} = 1{,}958$$

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